

Files, Terminals, and Your First Python Program

CS1 Week 3 — Monday, August 31, 2026

Computer Science I

The lesson in one sentence

The file manager and terminal are two views of the same filesystem; the computer follows exact instructions.

Predict. Try one small change. Read the result. Verify.

Two views, one location

File manager

- Visual folders and files
- Create and open items
- Save changes explicitly

Terminal

- Text commands
- Exact current location
- Ask it to list or inspect

Where am I? What is here?

```
ls  
cd GIT
```

- `ls` lists the current directory.
- `cd` changes the current directory.
- After creating a file, list again: the terminal will not guess.

Files have jobs and names

Extension	Kind	Use
.txt	plain text	notes or simple text
.md	Markdown	structured readable notes
.py	Python source	instructions Python can execute

A .py file is text on disk; it becomes a program when its contents and execution context are correct.

Your first tiny Python file

```
# A short note for another human  
print("hello world")
```

- # begins a comment: useful to humans, ignored by Python.
- print() is a function that sends text to output.

Literal machine, useful partner

- Python does not infer your intention.
- Names, punctuation, capitalization, and location matter.
- Once you learn the language, the machine becomes a resource instead of a mystery.

Case sensitivity is a rule

```
print("hello world")    # works  
Print("hello world")    # different name: error
```

Humans see the intended word. Python sees two different spellings.

Errors are information

- ① Read the error and locate the line.
- ② Predict one small correction.
- ③ Change one thing and run again.
- ④ Ask a person or tool to translate unfamiliar language.

An error costs a little time; the correction teaches you what the rule is.

Small tools from the lecture

- **Up arrow:** recall the previous terminal command.
- **Tab:** complete a name and check available choices.
- **Ctrl+S:** save the file before testing it.
- **File inspection:** look at the text before asking what it does.

Before Wednesday

- Save a tiny `.py` file with one comment and one `print()` line.
- Predict its output before running it.
- Wednesday: paired reasoning with the Week 3 material.
- Friday: show-and-tell/check-in on what worked and what you learned.

Be precise, be curious, and use errors as evidence.